

Practical - 5

Practical Title: Registration Form with Frontend Validation and User-Friendly Error Handling

Problem Definition: Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

Theory :

1. Are correct input types and attributes used?

Yes, correct input types and attributes are used. Text is used for names, email for email addresses, password for passwords, and the required attribute ensures that mandatory fields cannot be left empty.

2. Are validation errors displayed near the relevant fields?

No. Validation errors are currently displayed through alert messages rather than near the relevant input fields.

3. Is password strength checked?

Yes, password strength is checked using JavaScript validation with minimum length and character requirements.

4. Is the form accessible using keyboard and screen-reader-friendly labels?

Partially. Keyboard access is supported, but labels should be linked to their corresponding input fields using the for attribute for better screen-reader accessibility.

Validation:

```
// Name validation
if (!/^[A-Za-z\s]{3,50}$/.test(name)) {
    alert("🚫 Enter a valid name (letters and spaces only, at least 3 characters)!");
    return;
}

// Email validation
if (!/^[^\\s@]+@[^\\s@]+\\.([^\\s@]+)$/.test(emailValue)) {
    alert("✉ Enter a valid email address!");
    return;
}

// Mobile validation
if (!/^[0-9]{10}$/.test(mobileValue)) {
    alert("📞 Enter a valid 10-digit mobile number!");
    return;
}

// Gender validation
if (!genderValue) {
    alert("👤 Please select your gender!");
    return;
}

// Password validation
if (passwordValue.length < 6) {
    alert("🔒 Password must contain at least 6 characters!");
    return;
}
```

Implementation/ Output/Result

Registration Form

Full Name	ayush
Email	Enter Email
Mobile	78389380 ! Please fill in this field.
Password	
Confirm Password	
Date of Birth	08-10-2026
Gender	☒ Male ☐ Female ☐ Other
Course	B.Tech CE
Skills	☐ C ☐ C++ ☐ Java ☒ Python
Address	Enter Your Address

☐ I agree to the Terms & Conditions

GitHub Repository Link:

<https://github.com/ayushposhiya00-lab/Student-Hub/tree/main/Practical-4%265>

Learning Outcomes:

- Understand HTML form creation and different input types.
- Implement client-side form validation using JavaScript.
- Use Regular Expressions for validating user inputs.
- Implement password strength and confirmation checks.
- Handle validation errors and provide user-friendly messages.
- Understand basic form accessibility for keyboard and screen-reader users.

Faculty Remarks: (Keep this blank for faculty)